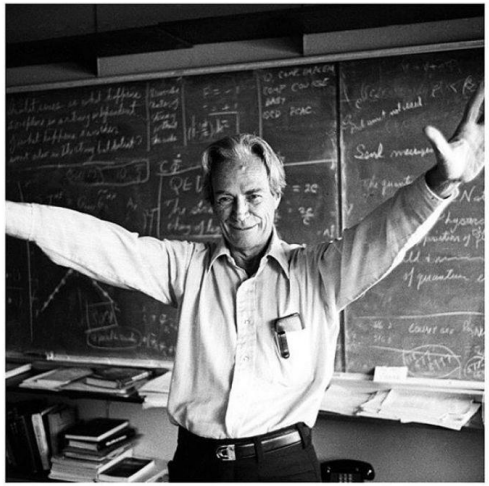
EART11200
Natural Scientists Toolkit
Maths and Physics



Professor Ann Webb

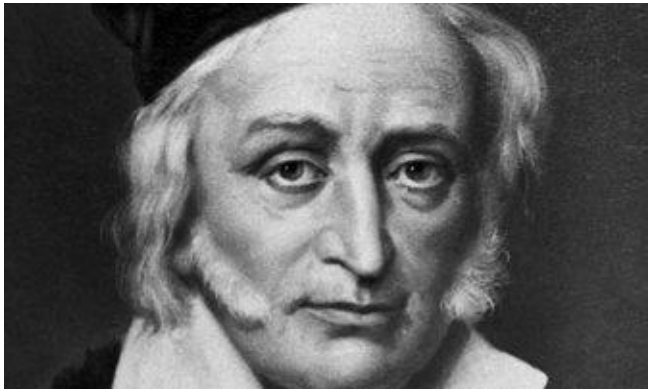
“To not know math is a severe limitation to understanding the world”

Richard P. Feynman



“There is nothing that living things do that cannot be understood from the point of view that they are made of atoms acting according to the laws of physics”

Richard P. Feynman



“Mathematics is the queen of science, and arithmetic the queen of mathematics”

Carl Friedrich Gauss

MOST WANTED II

Postgraduate and Professional Skills Needs in the Environment Sector

The 'Most Wanted II' report shows that UK environment sector desperately needs people trained in:

1. Modelling
2. Multi-disciplinarity
3. Data Management
4. Numeracy
5. Translating Research into practice
6. Fieldwork
7. Risk and Uncertainty
8. Taxonomy and Systematics
9. Soil Science
10. Environmental Epidermiology
11. Sustainability Science and Planning
12. Microbiology
13. Energy Supply
14. Food supply
15. Freshwater Science

More information on the 'Most Wanted II' report is available at:

Natural Environment Research Council website (Weblink)

Email: skills@nerc.ac.uk

Facebook page: www.facebook.com/lwec.org.uk

Follow us on Twitter: #lwec_uk

Why is maths important in our science?

- Geology and Environmental Sciences are not just qualitative science (describing fossils, or how pollution affects organisms).

- Like all *true* science they are quantitative.

- Usually we try to predict what will happen in the future, or explain what has happened in the past – often as a way of helping us make those future predictions.

Why is maths important in science 2

- Makes you more employable. Our Industrial Liaison panel were asked the following Qs:

  Q. What are the main skills you look for, or would be looking for in potential employees?

  • ***"Standard degree level skills e.g. numeracy, team working, presentation"*** *Robert Finney BP.*

  Q. Do you look for a reasonably high level of mathematics or quantitative skills in potential graduate employees?

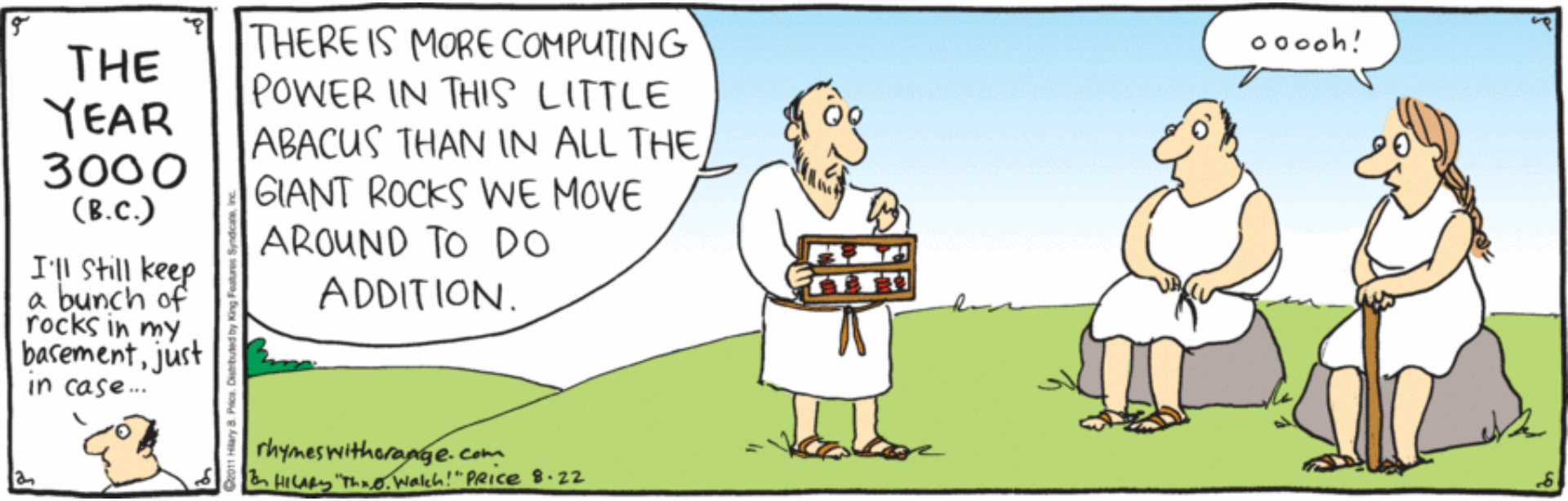
  • ***"Yes as a means to demonstrate general numeracy and also to enable both calculation of emissions, discharges, conversion into appropriate units, etc and impact modelling."*** *Shaun Robinson, Environment Agency*

  • ***"we are looking for people who can `do stuff', numerically",*** *Paul Field, Met Office*

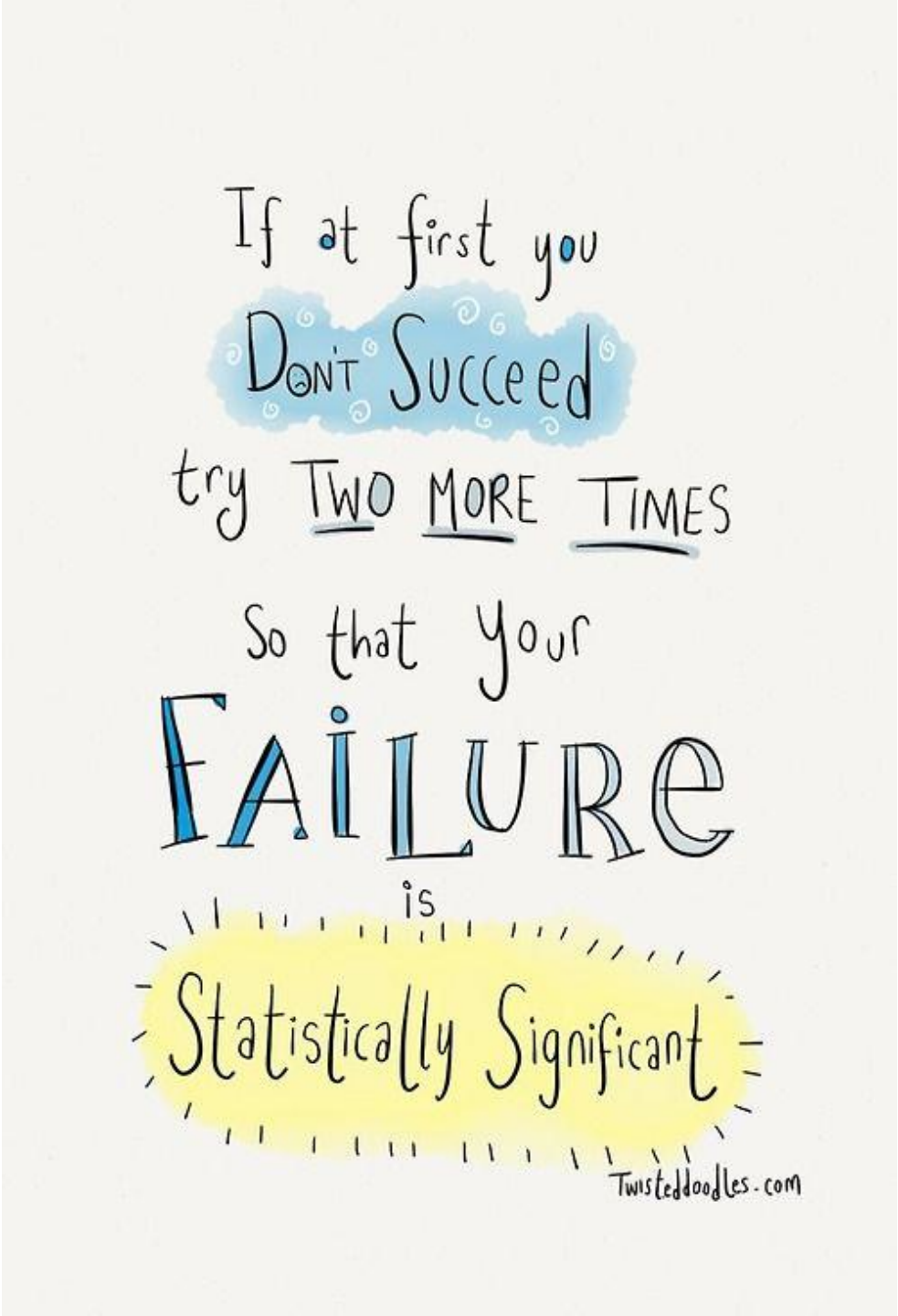
Starting points

- Use your critical faculty
  - e.g. mass of a lake 100m diameter: if you calculate that it weighs 50-80 kg, think does that make sense?
  - Does it fit in with what you know already?
- Converting units.
  - A fairly basic thing, but a common problem. So we will spend some time talking through it.
- You can all develop simple mathematical models of real world situations and use these to predict the future or past
  - This can be very useful, but also be aware of the degree of uncertainty in your predictions, either from uncertain input data, or oversimplified models

We will be providing you with the basic skills that make up your ‘toolkit’



Once you have the tools you will learn how to apply them in many different situations throughout your degree course.



- Look for hints in the question.
- Re-read the problem
- Look for the actual question (highlight it)
- Circle important data / information
- Double-check that you copied everything correctly
- Try drawing a picture – it is a great skill to develop